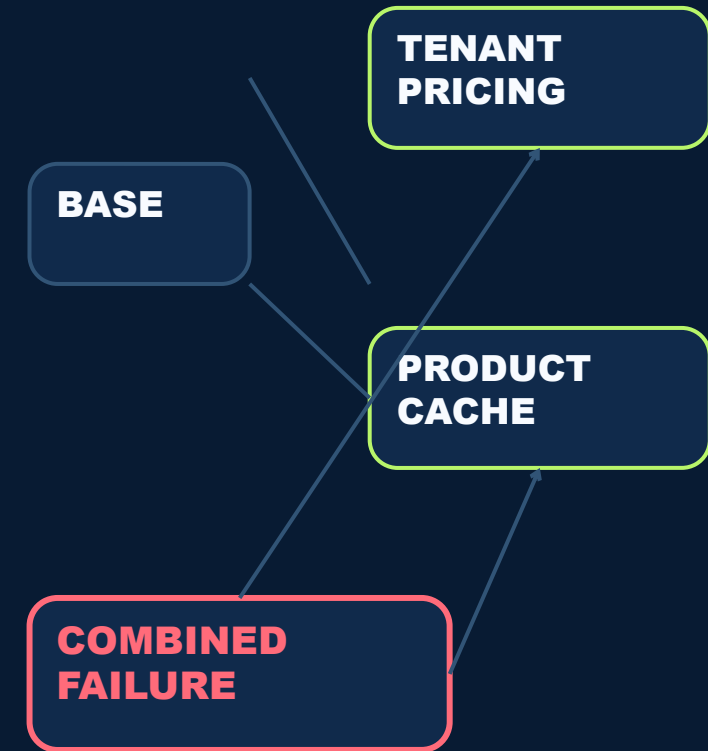


MergeWitness

A developer workflow for bugs that appear only when two green changes meet.



SYNTHETIC FIXTURE
Interaction witness verified

The failure hides behind green tests

Measured synthetic fixture. Original tests exit with code 0 in all four snapshots.

CHANGE A

Tenant-specific prices
Alpha: \$90 Beta: \$100

CHANGE B

Cache by product
Repeated requests avoid
source calls

- | | | |
|---|----------------------|--------|
| 1 | Alpha requests SKU-1 | \$90 |
| 2 | Cache stores SKU-1 | \$90 |
| 3 | Beta requests SKU-1 | \$90 × |

Expected for Beta

price[SKU-1] = 90

The ordinary suite can still pass because it checks each feature alone.

One invariant across four snapshots

Measured synthetic fixture. Three repetitions per snapshot; original tests exit with code 0 in all four.

Invariant: serving Alpha must not change Beta's price

Base

● Tests: exit 0

● Probe: 3× PASS

Change A

● Tests: exit 0

● Probe: 3× PASS

Change B

● Tests: exit 0

● Probe: 3× PASS

Combined

● Tests: exit 0

● **Probe: 3× FAIL**

Classification: interaction witness | Combined result: expected 100, observed 90

IBM Bob inside the investigation loop

Both relevant task summaries are recorded in bob_sessions/.

1

Task 1: original probe

fad890dd...bf59f |
0.953 Bobcoins

2

Audit derivatives

Independent
strengthened checks

3

Task 2: repair

Bob-authored nested-
Map repair

4

Task 2 evidence

d093059...6c74 |
0.637 Bobcoins

Recorded: 01-tenant-cache-probe-summary.png and 02-tenant-cache-repair-summary.png in bob_sessions/.

A repair must keep both features

Measured synthetic fixture. Final audit passed at 17ed2d8c...d94e.

Before

`cache[sku] = price`

Beta receives Alpha's cached price

?

Bob-authored repair

Nested Map cache

Tenant and SKU remain distinct

Frozen interaction probe

● VERIFIED PASS

Tenant prices remain distinct

● VERIFIED PASS

Cache remains effective

● VERIFIED PASS

A reproducible release decision

MergeWitness reports what the executed interaction probe found.

Run the laboratory

Inspect the input sequence and the observed result

Read the report

See commits, probe hash, and reproduction commands

Review the repair

Check the frozen probe and feature checks

A passing probe means this probe found no witness. It does not certify every merge as safe.